Republic of the Philippines
Commission on Higher Education
Region V (Bicol)
BICOL UNIVERSITY - POLANGUI
Polangui, Albay

CPE 26: MICROPROCESSOR

Microprocessor Laboratory Activities: Compilation of Labs 1 to 7

Submitted By :

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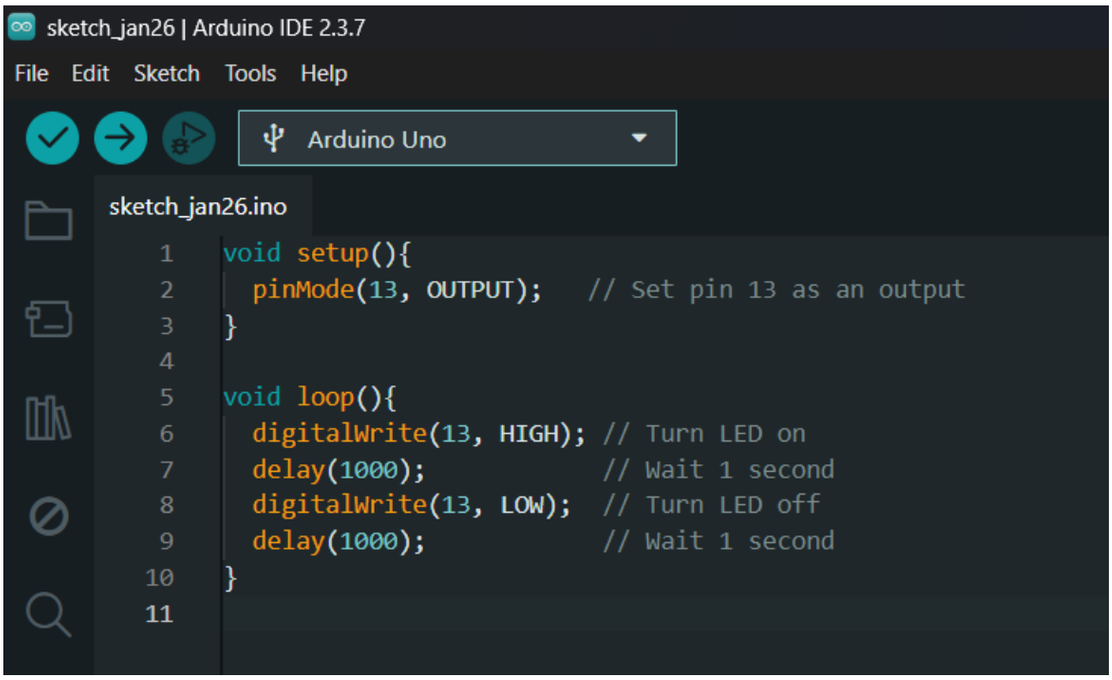
BSCpE – 3A

Submitted to :

Exequiel Mirasol Sabater

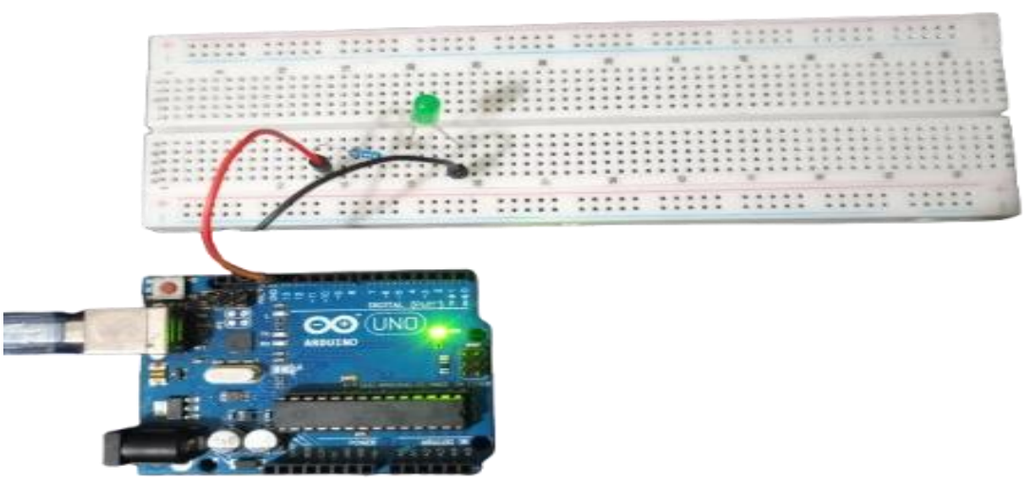
Date : March 16, 2026

Documentations For Lab 1:




```
void setup(){
  pinMode(13, OUTPUT);   // Set pin 13 as an output
}

void loop(){
  digitalWrite(13, HIGH); // Turn LED on
  delay(1000);            // Wait 1 second
  digitalWrite(13, LOW);  // Turn LED off
  delay(1000);            // Wait 1 second
}
```



Documentations For Lab 2:

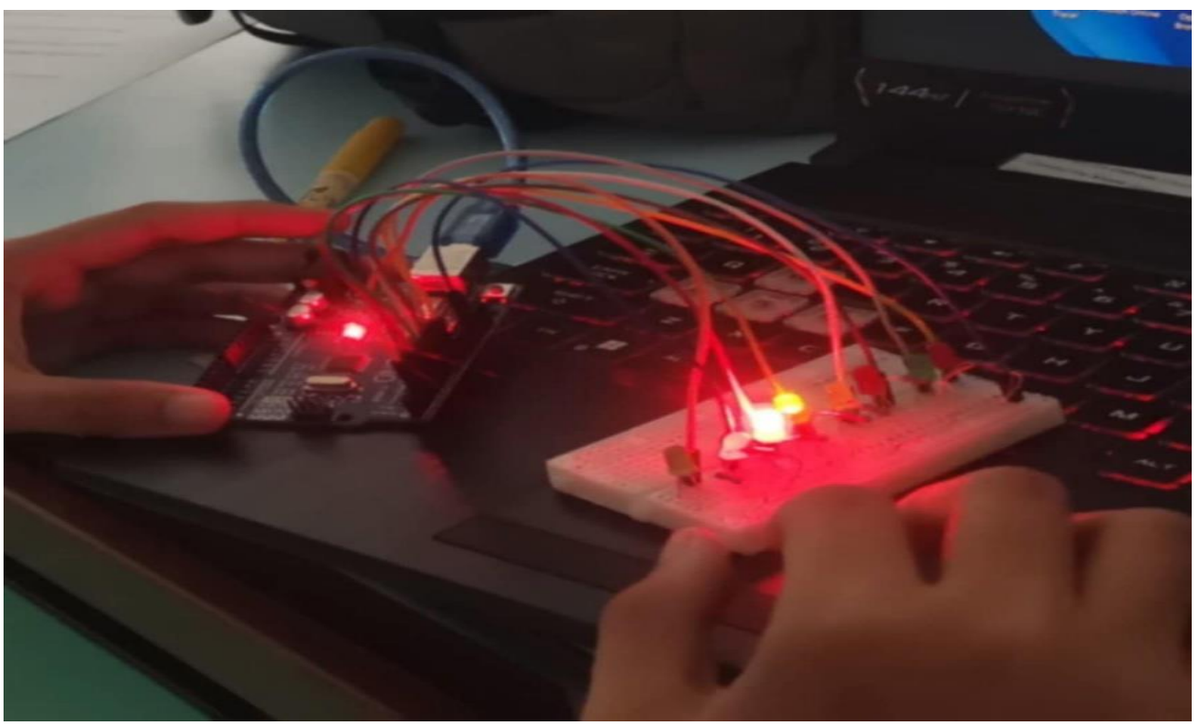
sketch_jan26 | Arduino IDE 2.3.7

File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
void setup(){
//Inititalize pins 2 through 9 as OUTPUT using a loop
    for (int pin = 2; pin <=9; pin++){
      pinMode(pin, OUTPUT);
}
}
void loop(){
//Pattern 1: The “Chase” (Left to Right)
    for(int pin = 2; pin<= 9; pin++){
      digitalWrite(pin,HIGH); // Turn LED on
      delay(100);
      digitalWrite(pin,LOW); // Turn LED off
}
//Pattern 2: The “Return” (Right to Left)
    for(int pin = 9; pin >=2; pin--){
      digitalWrite(pin,HIGH);
      delay(100);
      digitalWrite(pin,LOW);
}
}
```



Documentations For Lab 3:

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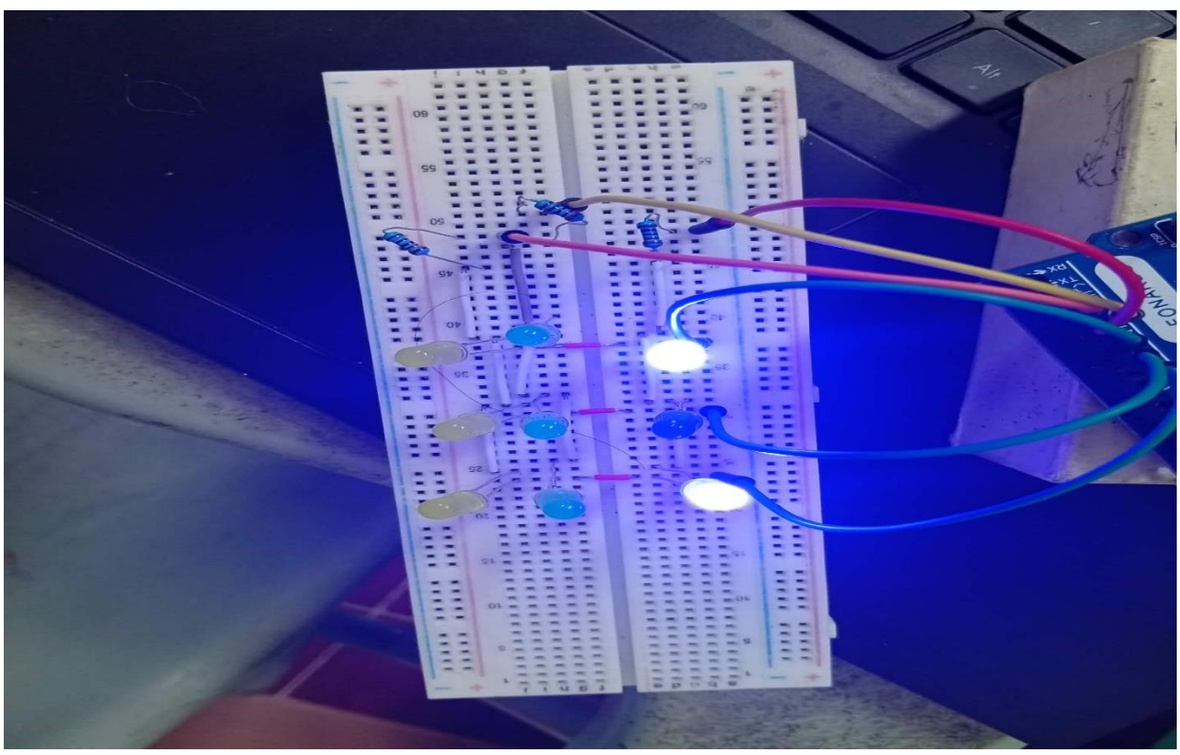
sketch_jan26.ino

```
// Pin definitions for rows and columns
int rowPins[] = {2, 3, 4};   // Rows connected to pins 2, 3, 4
int colPins[] = {5, 6, 7};   // Columns connected to pins 5, 6, 7
// Define the pattern to display on the matrix (1 = LED on, 0 = LED off)
byte pattern[3][3] = {
  {1, 0, 1},  // First row
  {0, 1, 0},  // Second row
  {1, 0, 1}   // Third row
};

void setup() {
  // Set all row pins as output
  for (int i = 0; i < 3; i++) {
    pinMode(rowPins[i], OUTPUT);
  }

  // Set all column pins as output
  for (int i = 0; i < 3; i++) {
    pinMode(colPins[i], OUTPUT);
  }
}
void loop() {
  // Loop through each row and column to display the pattern
  for (int row = 0; row < 3; row++) {
    // Turn on the current row
    digitalWrite(rowPins[row], HIGH);

    for (int col = 0; col < 3; col++) {
      // Turn on the LED in the current row and column if the pattern is 1
      if (pattern[row][col] == 1) {
        digitalWrite(colPins[col], HIGH);
      } else {
        digitalWrite(colPins[col], LOW);
      }
    }
    delay(500); // Delay to make the LED visible
    // Turn off the current row
    digitalWrite(rowPins[row], LOW);
  }
}
```



Documentations For Lab 5:

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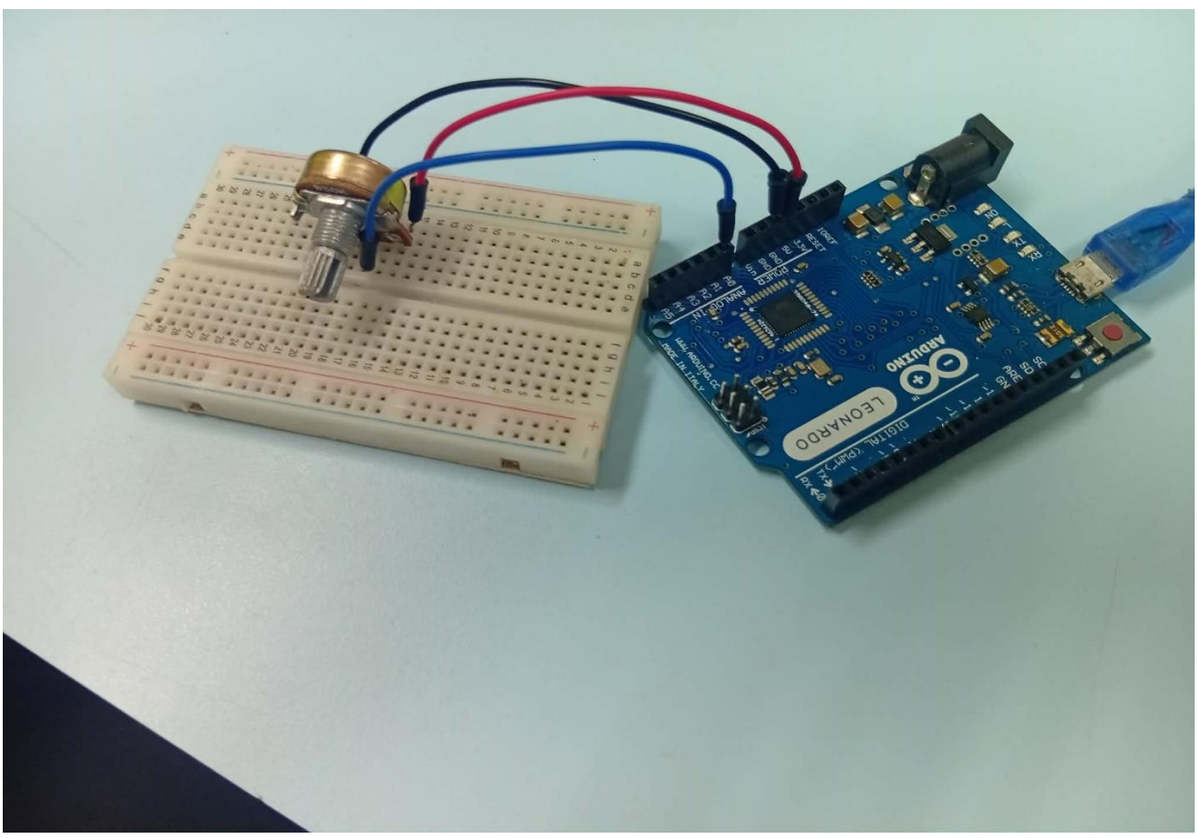
File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
/*
  Analog Input Lab
  Reads an analog input on pin 0, prints the result to the Serial Monitor.
*/
int potPin = A0;      // The middle pin of the potentiometer
int rawValue = 0;     // Variable to store the raw ADC value (0-1023)
float voltage = 0;    // Variable to store the calculated voltage

void setup() {
  Serial.begin(9600); // Open the serial port at 9600 bps
  Serial.println("Potentiometer Lab Started...");
}

void loop() {

  // Read the value from the sensor
  rawValue = analogRead(potPin);

  // Convert the raw value (0-1023) to voltage (0-5V)
  voltage = rawValue * (5.0 / 1023.0);

  // Print results to Serial Monitor
  Serial.print("Raw Value: ");
  Serial.print(rawValue);
  Serial.print(" | Voltage: ");
  Serial.print(voltage);
  Serial.println("V");

  delay(200); // Wait 200ms so the screen doesn't flicker too fast
}
```



Documentations For Lab 6:

sketch_jan26 | Arduino IDE 2.3.7

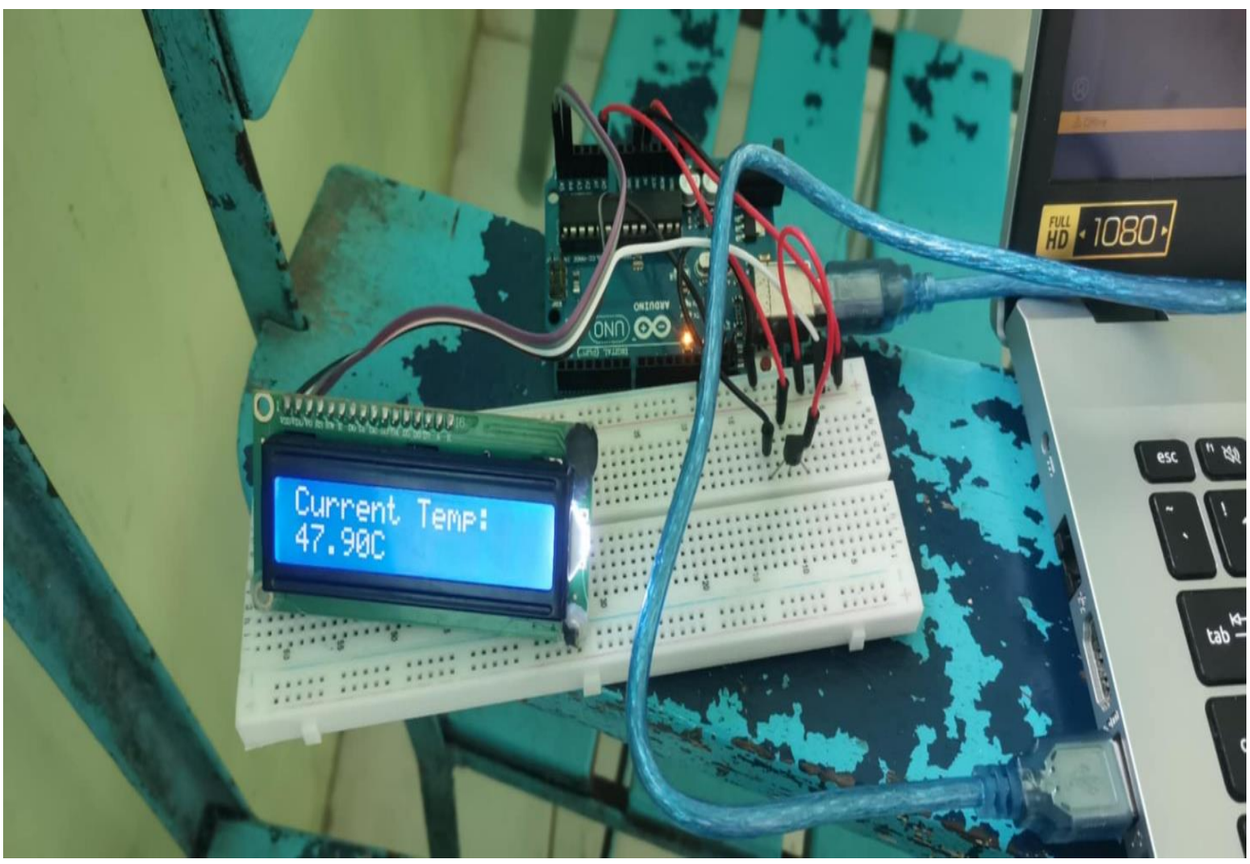
File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Initialize the LCD (Address 0x27 is common)
LiquidCrystal_I2C lcd(0x27, 16, 2);

const int sensorPin = A0;

void setup() {

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Temp Monitor");
  delay(2000);

  lcd.clear();
}

void loop() {

  // Read analog value (0-1023)
  int reading = analogRead(sensorPin);

  // Convert reading to voltage (for 5V Arduino)
  float voltage = reading * (5.0 / 1024.0);

  // LM35 output is 10mV per degree Celsius
  float tempC = voltage * 100.0;

  // UI Output
  lcd.setCursor(0, 0);
  lcd.print("Current Temp:");

  lcd.setCursor(0, 1);
  lcd.print(tempC);
  lcd.print(" C");

  delay(1000); // Update every second
}
```

`



Documentations For Lab 7:

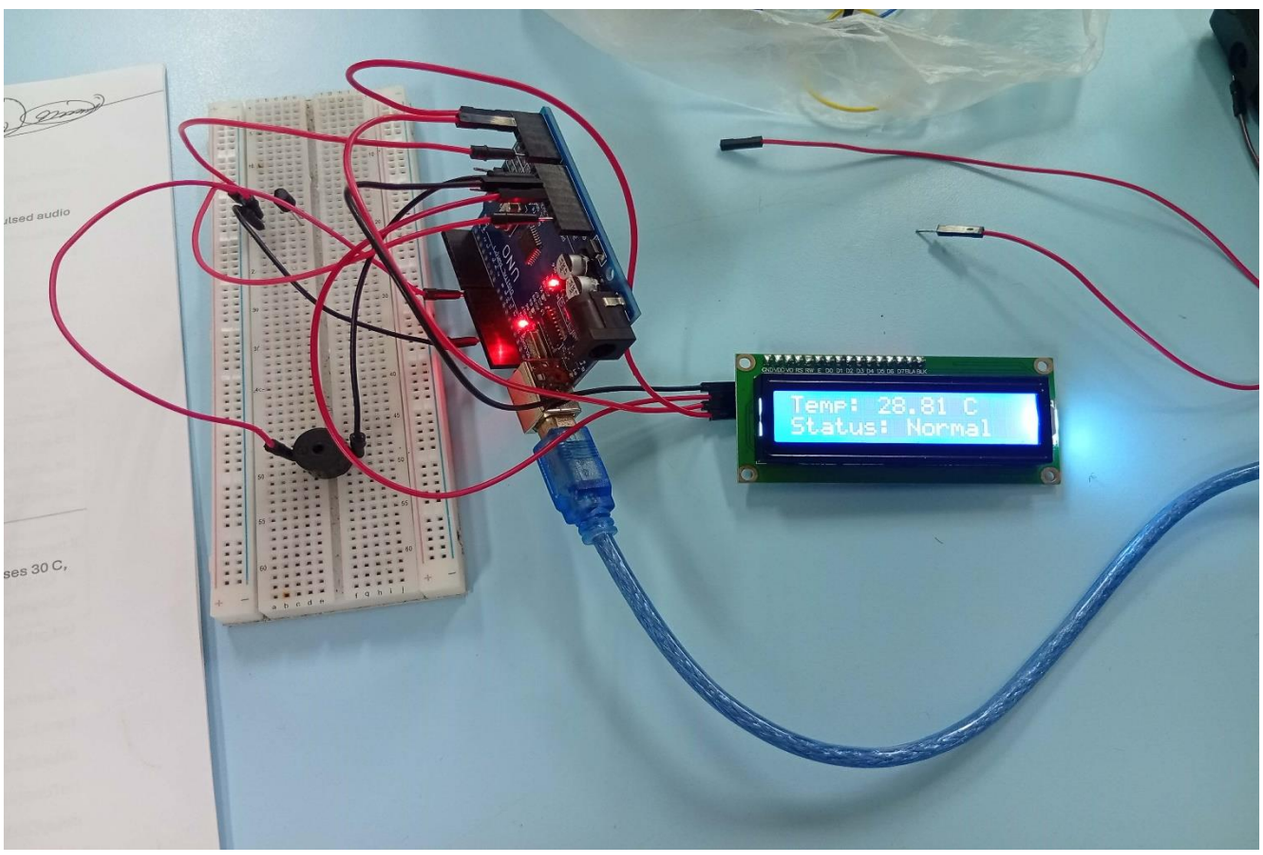
sketch_jan26 | Arduino IDE 2.3.7

File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

const int sensorPin = A0;
const int buzzerPin = 8;
const float threshold = 30.0; // Trigger temp in Celsius

void setup() {
  lcd.init();
  lcd.backlight();
  pinMode(buzzerPin, OUTPUT);
}

void loop() {

  int reading = analogRead(sensorPin);

  float voltage = reading * (5.0 / 1024.0);

  float tempC = voltage * 100.0;

  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(tempC);
```



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Microprocessor Laboratory Activity 1

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Microprocessor Laboratory Activity 2

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Microprocessor Laboratory Activity 3

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Microprocessor Laboratory Activity 4

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Microprocessor Laboratory Activity 5

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Microprocessor Laboratory Activity 6

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Microprocessor Laboratory Activity 7

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CpE Elec 2: COGNATE 2

COGNATE 2

Laboratory Activity 1

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CpE Elec 2: COGNATE 2

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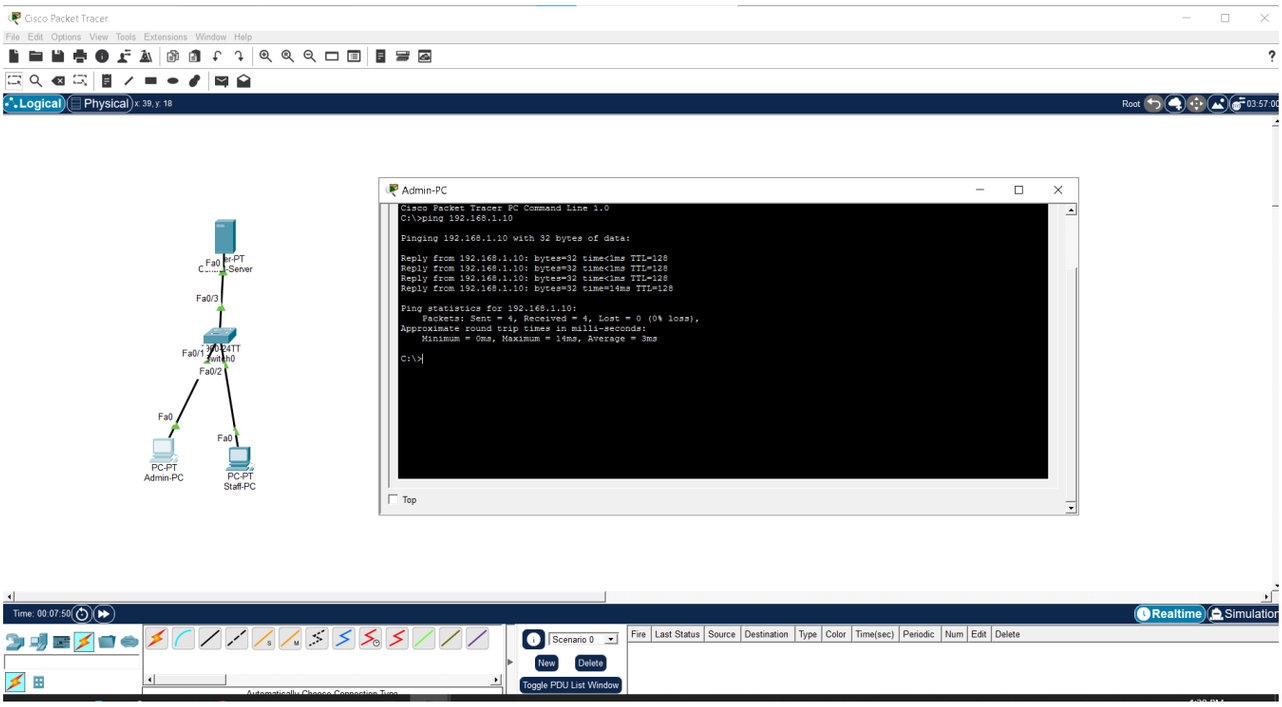
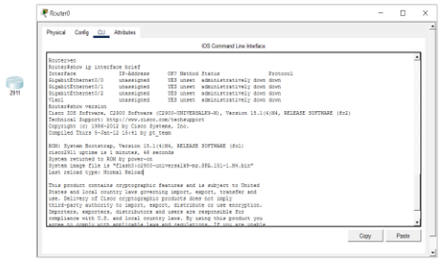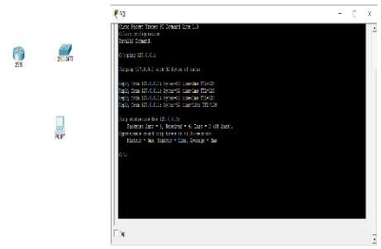
Documentations for Lab 2: